

Assignment Title: Introduction to Programming and Tools

Part 1: Introduction to Programming

1. What is programming?

- Programming is the process of writing a set of logical instructions that a computer follows **to** perform specific tasks.
- **p**rogramming is the method of communicating with computers using code to solve problems and automate tasks.
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2. Why is programming important in today's world?

- It makes our work faster by doing things automatically.
- It is used in schools, hospitals, banks, and many jobs.
- It solves problems and makes life easier with technology.
- Programming builds the smart tools we use every day.

3. List at least 3 real-life applications of programming.

- Google
- YouTube
- Whatsapp

4. What are programming languages? Mention at least 5 popular languages and one use case for each.

- Python – Used for data science and artificial intelligence
- JavaScript – Used for making interactive websites
- Java – Used for creating Android app
- C++ – Used in game development and high-performance apps
- SQL – Used for managing and

• **Part 2: Introduction to Python**

1. What is Python?

- Python is a simple, powerful programming language used to build software, automate tasks, and analyze data.

2. Who developed Python and when?

- **Python** was developed by Guido van Rossum and was first released in 1991.

3. What are the main features of Python?

- Easy to read and write
- Open-source and free
- High-level language
- Interpreted (no need to compile)
- **p**ortable (runs on any platform)
- Dynamically typed
- Large standard library
- Supports object-oriented and functional programming

4. Mention at least 3 areas where Python is widely used (e.g., data science, web development).

- Artificial Intelligence
- Machine Learning
- Data Science

5. What makes Python a beginner-friendly language?

- Simple and clean Syntax i.e print("hello, World") etc
- No need to declare Variables
- Fewer lines of Code
- Huge Community and Support
- Versatile USES

Part 3: Python IDEs

1. What is an IDE (Integrated Development Environment)?

- IDE is a software application for writing, testing, and debugging code. It offers tools like code editor, auto-complete, and error checking in one place.

2. List at least 3 popular IDEs or code editors used for Python development (e.g., PyCharm, VS Code, Jupyter Notebook).

- Visual studio code
- Jupyter notebook
- pycharm

3. Mention one advantage and one disadvantage of each IDE you listed.

1. PyCharm

- Advantage: Powerful features and smart code suggestions
- Disadvantage: Can be heavy and slow on low-end computers

2. VS Code

- Advantage: Lightweight, fast, and highly customizable
- Disadvantage: Needs extensions to add full Python support

Jupyter Notebook

- Advantage: Great for data analysis and visualizing results
- Disadvantage: Not ideal for building full software applications

Part 4: Git and GitHub

1. What is Git?

- **GIT is like a Time Machine for your code. it remember every change you make**

2. Why do developers use Git?

- **Track changes**
- **Backup code**
- **Work together**
- **Undo mistakes**
- **Organise work**

3. What is GitHub?

- **An Online platform to store, share, and collaborate on code using GiT.**

4. What is the difference between Git and GitHub?

- **Git is a tool for tracking code changes,**
- **while GitHub is a website for hosting and sharing Git repositories online.**

5. How does GitHub help in collaboration?

- **GitHub helps in collaboration by allowing multiple people to work on the same project, track changes, and merge their code easily.**

6. What is version control and why is it important in software development?

- **Version control is a system that tracks changes to code over time.**
- **It's important because it helps manage updates, fix mistakes, and collaborate with others safely in software development.**

Part 5: Introduction to Google Colab

1. What is Google Colab?

- Google Colab is a free online Jupyter notebook for writing and running Python code.

2. How is Google Colab different from other IDEs?

- Google Colab runs in the cloud, needs no setup, and supports free GPU for machine learning—unlike most IDEs that run locally on your computer.

3. What are its key features (e.g., free GPU access, cloud-based notebooks)?

- **Cloud-based** – No installation needed, runs in the browser
- **Free GPU/TPU access** – Great for machine learning and deep learning
- **Notebook interface** – Interactive coding with text, code, and output together
- **Auto-save to Google Drive** – Keeps your work safe
- **Easy sharing** – Just like sharing a Google Doc
- **Supports Python** – Especially for data science and AI tasks

4. What are the common use cases of Google Colab, especially in machine learning or data analysis?

- **Training machine learning models**
- **Data cleaning and analysis**
- **Building deep learning projects using TensorFlow or PyTorch**
- **Visualizing data with graphs and charts**
- **Running Python code without installing anything**

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